

Script Controlled ACL – Restrict Record Access Based on Field

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Requirement Analysis:

1. Introduction

The purpose of this project is to improve record-level security in ServiceNow. In a ServiceNow environment, different users may have different access requirements. Some records may contain sensitive or restricted information and should not be accessible to every user.

This project uses a **Script Controlled Access Control List (ACL)** to check a specific field value and decide whether a user should be allowed or denied access to a record.

2. Problem Identification

The existing access rules may allow users to access records that they are not authorized to view. Simple role-based permissions may not always be sufficient when access depends on the value of a particular record field.

Therefore, a dynamic field-based access mechanism is required.

3. Functional Requirements

The system should:

1. Identify the ServiceNow table that requires access restriction.
2. Identify or create a specific field for controlling access.

3. Store appropriate values in the field.
4. Create an ACL for the required table and operation.
5. Use an ACL script to check the field value.
6. Allow authorized users to access permitted records.
7. Deny unauthorized users from accessing restricted records.
8. Apply the access rule automatically whenever the record is accessed.
9. Support testing with different users and different field values.

4. Example

Suppose the field is:

Access Level

The field can contain:

- **Normal**
- **Restricted**

The access logic can be:

Access Level = Normal

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Allow Access

Access Level = Restricted

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Check User Authorization

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Authorized → Allow

Unauthorized → Deny

5. User Requirements

The users should be able to:

- Access records for which they have permission.
- View records according to their authorization.
- Work with normal records without unnecessary restrictions.

The system should prevent unauthorized users from accessing restricted records.

6. Security Requirements

The system should:

- Prevent unauthorized record access.
- Apply ACL rules consistently.
- Validate access at the record level.
- Ensure that restricted information is protected.
- Use appropriate roles and permissions along with the ACL logic.

7. Non-Functional Requirements

Security:

The system must protect restricted records from unauthorized access.

Reliability:

The ACL should provide consistent access decisions.

Performance:

The access check should execute efficiently without unnecessary processing.

Maintainability:

The ACL script should be simple and easy to update.

Scalability:

The solution should be adaptable to additional fields, users, or access rules if required.

8. Input Requirements

The system requires:

- ServiceNow record
- User information
- User role/authorization
- Access-control field
- Field value

9. Processing Requirements

The ACL should:

1. Receive the access request.
2. Identify the current record.
3. Check the required field value.
4. Check the user's authorization where applicable.
5. Determine whether access should be allowed or denied.

10. Output Requirements

The system should produce one of two outcomes:

- **Access Allowed** – The user can access the record.
- **Access Denied** – The user cannot access the record.

11. Testing Requirements

The project should test:

Test Case	Field Value	User Authorization	Expected Result
1	Normal	Authorized	Allow
2	Normal	Unauthorized	Based on defined ACL rule
3	Restricted	Authorized	Allow
4	Restricted	Unauthorized	Deny
5	Field value changed	Re-check authorization	Updated access decision

12. Main Objective

To design and implement a Script Controlled ACL in ServiceNow that dynamically restricts record access based on a specific field value and user authorization, thereby improving record-level security.